INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — Pseudo Code Execution & Logic Tracing

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — Pseudo Code Execution & Logic Tracing

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/pseudo-code/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 04_Pseudo_Code
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4, quiz-5, quiz-6

📋 Section 1: Rules to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Bitwise Operators: Bitwise AND (&) requires both bits 1; Bitwise OR (|) requires at least one bit 1; Bitwise XOR (^) yields 1 if bits differ.
- XOR Axioms: x ^ x = 0; x ^ 0 = x; XOR is commutative and associative.
- Bit Shifts: x << k multiplies by 2^k; x >> k divides by 2^k (integer floor division).
- Pre/Post Increment: ++x increments first and returns new value; x++ returns old value and increments afterwards.
- Operator Precedence: Parentheses > Postfix > Prefix/Unary > Multiplicative (*, /, %) > Additive (+, -) > Relational > Equality > Bitwise (&, ^, |) > Logical (&&, ||) > Assignment.

⚡ Section 2: How to Solve Quickly

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🚀 Section 3: Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Construct a Variable Trace Table with columns for loop index, variables, and conditions.
- Look for mathematical shortcuts: e.g. a loop summing 1 to N evaluates to N(N+1)/2 without step-by-step iteration.

📚 Section 4: Practice Questions, Options & Step-by-Step Solutions

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
What will be the output of the following pseudocode for i=140?

integer fun(int i)

if((i%2)!=0)

return i;

else

return fun(fun(i=1));

End function fun()

Options:
- **(A)** `1`
- **(B)** `2`
- **(C)** `3`
- **(D)** `Error`

Correct Answer: (A) `1`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Explanation:


In the first step i=140 is passed through the function fun(), and if the condition becomes false
when it is checked so else will get executed.


When else gets executed, function fun() is called again and i=1 is passed and now condition in IF
gets true and 'i' whose value is 1 gets returned.
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:

include<stdio.h>

int main()

{

int x=2,y=0,z=3;

x>y ?(printf("%d", z)):(return z);

}

Options:
- **(A)** `2`
- **(B)** `3`
- **(C)** `Error`
- **(D)** `11`

Correct Answer: (C) `Error`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Explanation:


The operator ?: works as follows expr1 is evaluated first if it is true expr2 is evaluated otherwise expr3 is evaluated. hence in expressions the return statement can not be used in C-language. The return statement is used for returning from a function , you can't use inside ternary operator
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:
What will be the output of below code?

Integer a, b, c, d, e

Set a=50 , b=3, c=3 e=0

while(c>0)

d=a mod b

e= e + d + a

c= c - 1

End while

Print e

Options:
- **(A)** `165`
- **(B)** `160`
- **(C)** `170`
- **(D)** `Error`

Correct Answer: (B) `160`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
#include&lt;iostream&gt;

using namespace std;

int main()

{

    int a, b, c, d, e;

    a=50 , b=3, c=3, e=0;

    while(c&gt;0)

    {

        d=a % b;

        e= e + d + a;

        c= c - 1;

    }

    cout&lt;&lt;e;

    return 0;

}
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:

for (i = 1; i <= 6; i++)
for (j = i; j < 6; j++)
Print blank space

for (k = 1; k < (i * 2); k++)
Print *
End for
Line break
End for

Options:
- **(A)** `* / *** / ***** / ****** / *******` (5 rows)
- **(B)** `* / ** / *** / ***** / ******* / *********` (6 rows non-uniform)
- **(C)** `* / *** / ***** / ******* / ********* / ***********` (6 rows with odd stars: 1, 3, 5, 7, 9, 11)
- **(D)** `None of the above`

Correct Answer: (C) `* / *** / ***** / ******* / ********* / ***********` (6 rows with odd stars: 1, 3, 5, 7, 9, 11)

PrepInsta Detailed Explanation & Mathematical Derivation:

```
The outer loop runs for i = 1 to 6 (representing 6 rows).
In each row i:
1. The first inner loop prints (6 - i) leading blank spaces.
2. The second inner loop prints * for k = 1 to (i * 2 - 1), which generates exactly (2i - 1) stars
(the odd numbers):
   - Row 1 (i=1): 2(1)-1 = 1 star
   - Row 2 (i=2): 2(2)-1 = 3 stars
   - Row 3 (i=3): 2(3)-1 = 5 stars
   - Row 4 (i=4): 2(4)-1 = 7 stars
   - Row 5 (i=5): 2(5)-1 = 9 stars
   - Row 6 (i=6): 2(6)-1 = 11 stars
3. A line break is printed at the end of each row.
Thus, this prints a centered pyramid of 6 rows with odd star counts from 1 to 11.
Option (C) is the exact output.
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
What will be the output of the following algorithm?

Start

Declare a=0, I and b

for I =0 to 4

Increment a by 1

if I = 3 then

print hello

get out of the loop

End if

End for

print a

Options:
- **(A)** `Hello4`
- **(B)** `Hello`
- **(C)** `Hello3`
- **(D)** `Error`

Correct Answer: (A) `Hello4`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Explanation:


Inside the loop with a value of I from 0 to 4, it will increment the value of 4 times that will be equal to 1, 2, 3, and 4.


but first inside the for loop hello will be printed when the value of I = 3


and at last, a will be printed i.e 4so the output will be hello4
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
What will be the output of below code?

#include<stdio.h>

int

f (int n)

{

static int a = 0;

if (n <= 0)

```
{

  return 1;

}
```

if (n > 3)

```
{

  a = n;

  return f (n - 2) + 2;

}
```

return f (n - 1) + a;

}

int main ()

{

printf ("Result: %d", f (5));

return 0;

}

Options:
- **(A)** `20`
- **(B)** `18`
- **(C)** `16`
- **(D)** `Error`

Correct Answer: (B) `18`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Explanation:


There are no errors in the given code, the output produced by the code is 18
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:
What will be the output of below code?

integer a

if((a mod 10) IS EQUAL TO 0)

a=a*2

else if((a mod 5) IS EQUAL TO 0)

a=a/5

else

a=a-1

end if

Options:

- **(A)** `5, 15`
- **(B)** `15, 5`
- **(C)** `25, 5`
- **(D)** `Compiler Error`

Correct Answer: (A) `5, 15`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
#include&lt;iostream&gt;

using namespace std;

int main()

{

int a =16;

if((a % 10)== 0)

a=a*2;

else if((a % 5 )== 0)

a=a/5;

else

a=a-1;

cout&lt;&lt;a;

return 0;

}
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:
integer a,b;

Set a=2; b=50;

while(b>0)

a = b%2 + a;

if(a MOD 2 Is Equal To 0)

Printf a

else

Print b-1

b = b/5

a=a+1

End while

Options:
- **(A)** `2 9 4`
- **(B)** `4 9 2`
- **(C)** `8 18 4`
- **(D)** `Error`

Correct Answer: (A) `2 9 4`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Explanation:

Step 1:

Condition true for the while loop

a = 2

Condition true for if

Value of an i.e 2 will get printed

Now b=10

and a=3

Step 2:

Condition true for the while loop

a = 3

Condition false for if, else will get printed

i.e. 9 will get printed

Now b=2

and a=4

Step 3:

Condition true for the while loop

a = 4

Condition true for if
```

```
Value of an i.e 4 will get printed


Now b=0


and a=5


Step 4:


Condition false for while. Exit
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:

Integer array1[10] = {2, 3, 56, 34}

Integer k, a, j, n

Set a = 3, n = 4

for(each k from 0 to n-1)

Set array1[n] = array1[0]

for(each j from 0 to n-1)

Set array1[j] = array1[j+1]

End for

End for

for(each k from 0 to n-1)

Print array1[k]

End for

Options:

- **(A)** `2 3 56 34`
- **(B)** `12 13 56 34`
- **(C)** `3 4 12 13`
- **(D)** `None of the mentioned`

Correct Answer: (A) `2 3 56 34`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
#include&lt;iostream&gt;

using namespace std;

int main()

{

int array1[10] = {2, 3, 56, 34};

int k, a, j, n;

a = 3, n = 4;

for(k=0 ;k&lt;=n-1;k++)

{

array1[n] = array1[0];

for(j=0;j&lt;=n-1;j++)

array1[j] = array1[j+1];

}

for(k=0;k&lt;=n-1;k++)

cout&lt;&lt;array1[k]&lt;&lt;" ";

return 0;

}
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:

What will be the output of the following pseudocode?

Integer x,y,z

Set x=3

Set y=90

while(y is greater than 0);

y=y/3

x=x+6

c=x+y

while(c is greater than 30):

if(c mod 3 is equals to 0):

Write x

else:

Write y

c=c/5

Write c

Options:

- **(A)** `9 18 15`
- **(B)** `6 33 9`
- **(C)** `9 33 6`
- **(D)** `None of the mentioned`

Correct Answer: (C) `9 33 6`

PrepInsta Detailed Explanation & Mathematical Derivation:

This pseudocode performs some simple mathematical operations until the value of y is greater than 0 and c is greater than 30 and prints the value of x, y, and z.

below is the Python code for this pseudo code:

```
x=3

x=3

y=90

while(y&gt;0):

y=y//3

x=x+6

c=x+y

while(c&gt;30):

if(c % 3 == 0):

print(x)

else:

print (y)

c=c//5

print(c)
```